Rationale-Augmented LLM Training for Targeted Hate Speech Detection in Vietnamese Social Media

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***Abstract*—Hate speech detection on Vietnamese social media remains challenging due to tonal ambiguity of the language, pervasive teencode, and culturally encoded implicit attacks that evade surface-level classifiers. Existing approaches largely ignore *why* content is hateful (the underlying stereotypes, harm mechanisms, and attack targets), limiting their ability to discriminate between offensive insults and dehumanizing hate. To address this, we adapt the HARE rationale-augmented training framework to Vietnamese targeted hate speech detection on ViTHSD, a 10,001-comment corpus with 11 multi-label target–severity classes. A two-stage QLoRA pipeline fine-tunes Qwen2.5-3B-Instruct: Stage 1 trains on 7,540 oversampled examples; Stage 2 continues on 1,221 LLM-generated, human-reviewed Chain-of-Thought rationale tuples encoding target identification, harm-mechanism reasoning, and severity justification. Multi-seed evaluation shows HARE achieves the best F1-Macro among compared models (35.93±0.63%) and reveals an asymmetric boundary redistribution, with disproportionate Offensive-level gains (+6.84 avg F1) alongside modest Hate-level improvement (+2.77 avg F1), not previously characterized in Vietnamese hate speech literature. These findings demonstrate that structured rationale supervision can improve fine-grained severity discrimination in low-resource, culturally complex settings, while highlighting the need for further disentangling rationale content from data-exposure effects.**

***Index Terms*—hate speech detection, Vietnamese NLP, rationale-augmented training, QLoRA, ViTHSD, Qwen2.5**

I. Introduction

Vietnamese social media platforms, including Facebook, YouTube, and TikTok, have become primary channels for public discourse while simultaneously enabling a surge in toxic and hateful content. The Vietnamese language presents unique moderation challenges: it employs a syllabic-based script with ambiguous word boundaries [16], is tonal with context-dependent meaning, and users frequently encode hate in cultural slang (e.g., *"ba que," "bò đỏ"*, English: "three-sticks [traitor]," "red cow [communist]"), teencode, sarcasm, and wordplay that evade keyword-based filters [17], [18]. Three compounding limitations affect existing automated approaches: (1) *lack of implicit-meaning modeling*, where classifiers operate on surface tokens without capturing the underlying stereotypes or harm mechanisms that distinguish Offensive from Hate content [12]; (2) *inability to detect implicit hate*, where surface-level models miss attacks expressed through cultural metaphor or sarcasm [14]; (3) *neglect of attack targets*, as prior work focuses on severity while overlooking who is attacked (an individual, a group, or a political entity), information essential for calibrating moderation responses [13]. A further difficulty is the *Offensive–Hate boundary*: implicit expressions often occupy the ambiguous space between insult and dehumanizing attack, causing boundary instability even when target identification is correct.

We address these challenges on ViTHSD [2], a dataset of 10,001 Vietnamese social media comments annotated with composite *[Target#Level]* labels, where Target $\in$ {Individuals, Groups, Religion, Race, Politics} and Level $\in$ {Offensive, Hate}, plus a Normal label, yielding 11 classes in a multi-label setting where each comment may carry multiple target–level pairs. We adapt the HARE framework [9] to this Vietnamese targeted task by generating 1,744 Chain-of-Thought (CoT) rationale tuples via Gemini 2.0 Flash [6] and training Qwen2.5-3B-Instruct via two-stage QLoRA [11].

This paper makes three contributions:

- **Rationale-Augmented Training for Vietnamese Targeted Hate Speech with Parse-Failure Analysis.** Under F1-Macro, HARE achieves the best score among compared models (Table II), i.e., baselines and Vanilla Stage 1-only (35.93±0.63%, pooled bootstrap $p<0.001$), and the best parseable-only F1-Micro (46.71% vs. 44.58%, $p=0.004$). The aggregate Vanilla F1-Micro advantage is an artifact of differential parse-failure rates (∼36% vs. ∼59%). Multi-seed ablation shows a modest advantage of rationale *semantic structure* over shuffled controls (+1.11pp mean F1-Macro); however, Plain-Continuation (same data, no rationales) achieves higher F1-Macro (37.05%), indicating the Stage 2 benefit partly reflects data exposure.
- **Hate/Offensive Boundary Redistribution.** Confirmed across three seeds, Stage 2 rationale training consistently shifts predictions toward Offensive-level labels (+6.84 avg F1) with modest Hate-level gains (+2.77 avg F1), an asymmetric improvement invisible in aggregate metrics and not previously characterized in Vietnamese hate speech literature.
- **Methodological Finding: Parse-Failure Metric Artifacts.** We show that differential parse-failure rates (∼36% vs. ∼59%) can reverse model rankings when unparseable outputs default to a majority class, and recommend parseable-only evaluation as standard practice for generative classifiers.

Sections II–IV present background and methodology; Sections V–VII report experiments; Section VIII concludes.

II. Related Work

Vietnamese NLP and Hate Speech. Pre-trained Transformer models for Vietnamese include PhoBERT [3], PhoGPT [20], and multilingual Qwen2.5 [4]. Early Vietnamese

hate speech detection studies leveraged the VLSP shared task dataset, where hybrid neural models and ensemble approaches showed effective performance [23]. ViHSD [1] later provided the first large-scale Vietnamese hate speech dataset (33,400 comments; Clean/Offensive/Hate), while ViTHSD [2] adds five target categories with multi-label severity pairs, reporting ViSoBERT at F1-Micro 51.12%. ViHOS [21] provides span-level rationale annotation; our approach uses document-level CoT rationales applied only during training.

Multilingual and Cross-Lingual Hate Speech Detection. Beyond Vietnamese, hate speech detection has been studied across diverse languages including Arabic, Hindi, and code-mixed settings, where shared challenges include cultural context dependence and implicit expression [17]. Multilingual Transformer models such as XLM-R and mBERT have shown cross-lingual transfer potential [18], but Vietnamese, a tonal, syllabic language with pervasive teencode, remains underexplored in cross-lingual evaluation.

Explainable Detection and Parameter-Efficient Fine-Tuning. The HARE framework [9] uses LLM-generated step-by-step rationales to improve both interpretability and classification. HateXplain [12] and Social Bias Frames [13] motivate target-aware explainable detection. Chain-of-Thought prompting [10] decomposes complex reasoning into interpretable chains. QLoRA [11] enables adapting large models on consumer hardware, making CoT-enhanced training feasible in resource-constrained settings.

Offensive–Hate Boundary. Davidson et al. [22] first documented systematic Offensive→Hate confusion; SemEval-2019 Task 5 [19] reports analogous instability. ElSherief et al. [14] show implicit hate occupies a low-confidence continuum. In Vietnamese, cultural slang and indirect reference compound this challenge; no prior study has tested whether training-time rationale supervision can systematically shift this boundary.

III. Methodology

This section first formalizes the task, then details model selection, the two-stage training pipeline, and the inference procedure.

A. Task Definition

Given a Vietnamese social media comment c, the task is multi-label classification: predict the label subset $Y \subseteq \mathcal{L}$, where $\mathcal{L} = \{\textit{Normal}\} \cup \{T\#L \mid T \in \mathcal{T},\ L \in \{\textit{Offensive}, \textit{Hate}\}\}$ with target set $\mathcal{T} = \{\textit{Individuals, Groups, Religion, Race, Politics}\}$, yielding $|\mathcal{L}|$=11. A comment may carry multiple target–level pairs simultaneously; a one-level-per-target constraint retains only the higher severity (Hate > Offensive) when both are predicted for the same target.

Our approach follows a two-phase pipeline (Fig. 1): (1) **Annotation**: generating rationales via a commercial LLM; and (2) **Training**: fine-tuning Qwen2.5-3B-Instruct in two stages. Unlike the original HARE [9], which extracts rationales at both training and inference time, our adaptation employs rationale supervision exclusively during training; inference produces only label predictions.

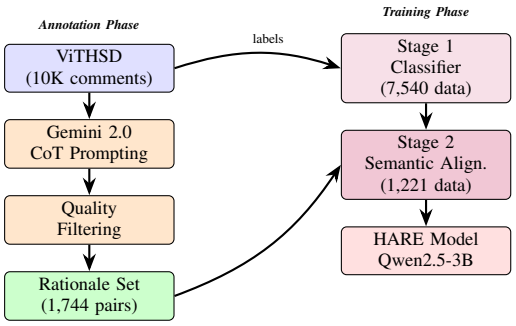

Figure 1. Two-phase pipeline: annotation (left) and two-stage training (right).

B. Model Selection

We select **Qwen2.5-3B-Instruct** [4]: (1) at 3B parameters it fits on a single T4 GPU (16 GB) under 4-bit quantization; (2) it performs competitively on Vietnamese benchmarks among sub-10B models vs. Llama 2-7B [7] and Gemma-7B [8]; and (3) its instruction-following capability supports structured output natively. The Qwen2.5 BPE tokenizer processes Vietnamese at the syllable level without word segmentation (unlike PhoBERT [16]), providing robustness to teencode and diacriticless input.

C. Two-Stage QLoRA Fine-Tuning

Both stages use QLoRA [11] with 4-bit NF4 quantization, LoRA [15] rank r=8, α=16, dropout 0.1, targeting all attention and feed-forward projection layers. Input uses the Qwen chat template.

Stage 1 (Classification): The model learns to predict multi-label *[Target#Level]* from the ViTHSD training set. We oversample labels with fewer than 200 training examples via random sampling with replacement, adding 540 additional samples concentrated in Politics, Race, and Religion, yielding 7,540 training examples used by all baselines. A post-processing step enforces one-level-per-target by preferring Hate over Offensive. Hyperparameters: 2 epochs, batch 4, lr 2×10^{-4}, max length 512.

Stage 2 (Rationale-Augmented Continuation): Starting from the Stage 1 checkpoint, training continues on 1,221 rationale-enriched training tuples, the training subset of the 1,744 annotated pairs (Sec. IV). Inputs are augmented with implied-statement and rationale fields, teaching the model to associate implicit expressions with their true semantic intent. Hyperparameters identical to Stage 1. All 1,221 Stage 2 examples overlap with Stage 1; these comments are seen twice (plain, then rationale-enriched), which may contribute to memorization.

D. Inference

The model generates free-text output (max 50 tokens) parsed via string matching against the 11-label vocabulary. All detected label strings are accepted without a confidence threshold (multi-label output accepts any subset of the 11

labels detected in the generated text), subject to a one-level-per-target constraint that prefers Hate over Offensive when both are detected for the same target. If parsing fails (at rates that vary substantially by model and training stage, Sec. VI-B), the comment defaults to *Normal*, inflating Normal-class precision and suppressing minority-label recall.

IV. Rationale Generation, Filtering, and Selection Bias

We employ Gemini 2.0 Flash [6] as the automated annotator. Through four prompt iterations (mean annotator scores: 40, 67, 89, 93 out of 100), we converged on a structured two-prompt pipeline comprising: (a) an *implied-statement prompt* that extracts noun-phrase stereotypes constrained to seven predefined categories (competence, morality, gender, ethnicity, class, religion/politics, appearance) with explicit negative examples to prevent speaker-action paraphrases; and (b) a *rationale prompt* enforcing a strict four-step CoT structure per comment: (1) *Nhận diện ngôn từ* (*Identify language cues*): identify offensive language cues; (2) *Cơ chế gây hại* (*Harm mechanism*): explain the harm mechanism; (3) *Liên kết implied statement* (*Link implied statement*): connect to the extracted stereotype; (4) *Phân biệt mức độ* (*Distinguish severity*): justify Hate vs. Offensive distinction. Outputs are enforced as structured JSON; scores reflect mean four-annotator holistic ratings on label consistency and cultural plausibility (0–100 scale).

Quality filtering. LLM-generated implied statements were produced for all 7,528 hate/offensive training instances and reviewed by four annotators (Vietnamese university students, ages 20–22, Ho Chi Minh City) for cultural appropriateness and label alignment. A 50-sample dual-reviewer Likert audit yielded Cohen's $\kappa = 0.72$ (substantial agreement). Remaining pairs passed an automatic consistency check: fed back into Gemini for label re-prediction, with mismatches discarded. Same-family verification is a known limitation (Sec. VII). The final set contains **1,744 tuples** (1,221 training, 321 test, 202 development; selection rate ≈23%); 5,784 (76.8%) were rejected, concentrated on ambiguous boundary cases and minority labels. Table I reports label distribution: Hate-level labels constitute 60.5% of label instances (1,582/2,617), confirming strong bias toward high-confidence Hate examples, a selection effect likely contributing to boundary redistribution (Sec. VI-B).

Selection bias analysis. The consistency filter produces a Hate/Offensive ratio reversal: training data is Offensive-dominant (66.3%), whereas the rationale set is Hate-dominant (60.5%, Δ=+26.8pp). This arises because Hate-level language provides clearer reasoning anchors; ambiguous Offensive cases are disproportionately rejected. The reversal connects to boundary redistribution (Sec. VI-B): Stage 2 training on a Hate-biased subset shifts the decision boundary.

V. Experimental Setup

Dataset and Metrics. ViTHSD [2] contains 10,001 Vietnamese comments with composite *[Target#Level]* labels across five targets (Individuals ∼60%, Groups ∼33%, Race ∼5%, Politics ∼4%, Religion <0.3%) and two severity levels per target, yielding 11 classes. All models train on the 7,540-sample oversampled training split and evaluate on the 1,800-sample test set. We report F1-Micro, F1-Macro, F1-Samples, Precision-Micro, and Recall-Micro via scikit-learn. HARE and Vanilla are evaluated under three random seeds (42, 123, 456); baselines are single-run. All experiments run on NVIDIA T4 GPUs (16 GB, Kaggle; ∼4.5 h per seed).[1] F1-Macro is our primary metric as it weights all 11 classes equally, making it more informative than F1-Micro for highly imbalanced settings.

Table I
Label distribution of the 1,744 accepted rationale tuples. Counts exceed 1,744 due to 718 multi-label comments.

Label	Count	%*
Individuals#Hate	878	50.3
Individuals#Offensive	566	32.5
Groups#Hate	451	25.9
Groups#Offensive	319	18.3
Politics#Hate	137	7.9
Race#Hate	110	6.3
Race#Offensive	88	5.0
Politics#Offensive	52	3.0
Religion#Offensive	10	0.6
Religion#Hate	6	0.3
Hate-level total	1,582	60.5†
Offensive-level total	1,035	39.5†

*% of 1,744 comments; sums >100 due to multi-label.
†% of 2,617 total label instances.

Baselines (all retrained under the same oversampled 7,540-sample protocol; raw text input, no word segmentation):

1) **BiGRU-LSTM-CNN**: Hybrid sequential model; batch 16, 3 epochs, lr 10^{-3}, max length 256.
2) **PhoBERT-base** [3]: Encoder-only; batch 16, 3 epochs, lr 2×10^{-5}. *Note*: pretrained on word-segmented text; unsegmented input is a preprocessing disadvantage not shared by Qwen2.5 or Flan-T5.
3) **Flan-T5-base** [5]: Encoder-decoder; batch 8, 3 epochs, lr 3×10^{-5}, max output 64 tokens.
4) **Qwen2.5-3B Vanilla**: Stage 1 QLoRA only (no rationale data).
5) **HARE (this work)**: Stage 1 + Stage 2 rationale continuation.

VI. Results and Analysis

This section presents multi-seed aggregate performance (Sec. VI-A), per-label boundary analysis (Sec. VI-B), and error characterization (Sec. VI-C) for the ViTHSD test set (1,800 samples).

A. Overall Performance

Table II presents multi-label performance (HARE and Vanilla averaged over three seeds). Under F1-Macro, HARE

[1]Code, rationale tuples (1,744 pairs), LoRA adapters, and parseable-only evaluation scripts will be released upon acceptance; a browser-based demo is available upon request.

Table II
MULTI-LABEL PERFORMANCE ON VITHSD TEST SET (%). HARE/VANILLA: MEAN±STD OVER SEEDS {42,123,456}. BASELINES (†): SINGLE-RUN. BOLD = BEST PER METRIC.

Metric	HARE ($\mu\pm\sigma$)	Vanilla ($\mu\pm\sigma$)	BiGRU-LSTM†	PhoBERT†	Flan-T5†
F1-Samples	60.67±0.90	**63.77**±0.68	58.49	58.01	50.74
F1-Micro	57.08±1.08	**60.04**±0.40	51.91	54.12	46.84
F1-Macro	**35.93**±0.63	32.75±0.53	31.90	25.86	13.11
Prec-Micro	52.82±2.18	**60.37**±2.04	41.98	56.20	48.10
Rec-Micro	62.15±0.75	59.79±1.55	**67.98**	53.10	45.20

Plain-Continuation (ablation control, Table VI) achieves 37.05% F1-Macro but is not a standalone model.

Table III
PER-SEED PARSE-FAILURE RATES AND PARSEABLE-ONLY F1-MICRO (%). BOOTSTRAP p-VALUES FROM 10,000-ITERATION PAIRED TESTS.

	Fail Rate (%)		Parse-only F1		
Seed	HARE	Van.	HARE	Van.	p
42	33	57	47.2	44.6	0.004
123	35	60	45.8	44.9	0.377
456	40	60	47.1	44.3	0.019
Pooled	36	59	46.7	44.6	0.004

achieves the highest score (35.93±0.63%), outperforming Vanilla (32.75±0.53%), BiGRU-LSTM-CNN (31.90%), PhoBERT (25.86%), and Flan-T5 (13.11%); a pooled bootstrap test confirms significance ($p<0.001$). HARE also leads Recall-Micro (62.15±0.75% vs. 59.79±1.55%). Vanilla achieves higher aggregate F1-Micro (60.04% vs. 57.08%) and Precision-Micro (60.37% vs. 52.82%); however, this advantage is a parse-failure artifact (see below). Most retrained baselines exceed the ViSoBERT reference (F1-Micro 51.12% [2]), reflecting oversampling and preprocessing differences. Despite the improvement of HARE, 9-class F1-Macro excluding Religion (43.91% HARE vs. 40.02% Vanilla) remains below deployment threshold. At 3B parameters, HARE requires ∼4.5 h/seed on a T4 GPU versus minutes for PhoBERT-base, a >50× compute premium for +10pp F1-Macro.

Parse-failure confound. HARE averages ∼36% parse failures versus ∼59% for Vanilla (Table III). All unparseable outputs default to *Normal*, inflating Normal-class precision and suppressing minority-label recall, disproportionately penalizing the model with *fewer* failures. The rate asymmetry likely reflects format reinforcement in Stage 2 through 1,221 additional structured label sequences. A paired bootstrap test on parseable outputs only (pooled across 5,400 test instances, 3 seeds × 1,800) reverses the F1-Micro ranking: HARE 46.71% vs. Vanilla 44.58% (p=0.004).

B. Per-Label Analysis and Boundary Redistribution

To identify where the aggregate F1-Macro improvement of HARE concentrates across the 11-class label space, we examine seed-averaged per-label F1 scores. Table IV presents these results. Excluding Religion (<30 test samples, 0% F1 for all models), **HARE outperforms Vanilla on 5 of 8 remaining labels**. The three high-frequency Offensive labels show large gains (Individuals#Offensive +13.36, Groups#Offensive +13.43, Race#Offensive +11.16), and Race#Hate reverses from parity to a HARE lead (+13.68). Individuals#Hate shows a modest gain (+0.82). Vanilla retains advantages on Groups#Hate (+3.16) and Politics#Offensive (+10.59); Politics#Hate is essentially tied (Δ<0.3). The **Boundary Redistribution Index**, the average HARE–Vanilla delta across non-religion labels, quantifies the asymmetry: $\Delta\overline{\text{Offens.}} = +6.84$ vs. $\Delta\overline{\text{Hate}} = +2.77$, with Offensive gains consistent across all three seeds. Possible mechanisms are discussed in Sec. VII.

Table IV
PER-LABEL F1 (%) ON VITHSD TEST SET. HARE/VANILLA: MEAN±STD OVER 3 SEEDS. Δ: HARE−VANILLA. BASELINES: SINGLE-RUN. †: <40 TEST SAMPLES. "–" = F1 = 0. BOLD = BEST MEAN PER ROW.

Label	HARE ($\mu\pm\sigma$)	Vanilla ($\mu\pm\sigma$)	Δ	BiGRU-LSTM†	PhoBERT†	Flan-T5†
Normal	75.7±0.5	**79.2**±0.9	−3.5	74.90	75.31	67.32
Individuals#Hate	**61.7**±1.0	60.9±1.0	+0.8	58.01	53.65	35.83
Individuals#Offensive	**42.0**±1.3	28.7±6.5	+13.4	38.60	7.73	–
Groups#Hate	51.9±2.5	**55.0**±1.1	−3.2	43.84	46.03	34.93
Groups#Offensive	**33.4**±2.7	20.0±5.7	+13.4	27.98	–	–
Religion#Hate†	–	–	–	–	–	–
Religion#Offensive†	–	–	–	–	–	–
Politics#Hate	51.4±2.6	**51.6**±1.1	−0.3	43.60	29.13	2.99
Politics#Offensive	4.6±4.0	15.2±5.8	−10.6	15.19	**21.18**	0.00
Race#Hate	**48.4**±5.3	34.7±16.4	+13.7	29.91	28.57	3.12
Race#Offensive	**26.1**±4.2	14.9±4.5	+11.2	18.89	22.92	–

Table V
PER-LABEL PRECISION (P) AND RECALL (R) DECOMPOSITION (%) FOR HARE VS. VANILLA, MEAN OVER 3 SEEDS. Δ: HARE−VANILLA.

	HARE		Vanilla		Δ (H−V)	
Label	P	R	P	R	ΔP	ΔR
Normal	86.8	67.3	73.4	86.0	+13.4	−18.7
Indiv#Hate	54.2	71.9	56.4	66.7	−2.3	+5.1
Indiv#Offensive	36.2	50.5	44.7	21.5	−8.5	**+29.0**
Groups#Hate	44.5	62.9	57.6	54.2	−13.2	+8.6
Groups#Offensive	26.4	46.1	31.5	15.0	−5.1	**+31.2**

Data scarcity. Religion scores 0% F1 across all models (<30 test samples; grayed rows in Table IV). Race and Politics remain low-resource with high variance; statistical conclusions apply primarily to Individuals and Groups (∼93% of test instances).

Precision–Recall decomposition. Table V decomposes the boundary redistribution. The Offensive gains of HARE are overwhelmingly *recall-driven*: Individuals#Offensive recall jumps from 21.5% to 50.5% (+29.0pp) while precision drops −8.5pp; Groups#Offensive shows the same pattern (+31.2/−5.1pp). Rationale training primarily lowers the prediction threshold for Offensive-level labels rather than improving discrimination per se.

This example illustrates how step (4) encodes the Hate/Offensive boundary signal driving the redistribution in Table V.

Post: *"ý thức của chị Nhung còn thua điểm lí của em"* (English: "Nhung's awareness is even worse than my physics grade.")
Gold labels: *Individuals#Hate.*
Implied: người khác là kém hiểu biết (English: "Others are ignorant").
Stage 2 Rationale: (1) *"thua điểm lí của em"* (English: "worse than my physics grade") is a sarcastic comparison to demean the moral value of the target (*Indiv* target); (2) *"ý thức ... còn thua"* (English: "awareness ... is even worse") implies the target has inferior conduct, worse than a poor academic score; (3) likening the character of a person to a grade is a direct attack on their dignity and honor; (4) mocking tone with intent to publicly attack an individual is consistent with *Hate*-level.

Figure 2. Stage 2 rationale example (ViTHSD training set).

Table VI
MULTI-SEED ABLATION (F1-MACRO, %). PLAIN-CONTINUATION INCLUDES NORMAL EXAMPLES.

Model	Seed 42	Seed 123	Seed 456	Mean
HARE (Stage 2)	35.37	36.61	35.81	**35.93**
Qwen Vanilla	32.26	33.32	32.67	32.75
Shuffled-Rationale	34.22	36.68	33.56	34.82
Plain-Continuation	35.12	37.48	38.55	37.05

C. Error Taxonomy

We categorize failure modes to guide future architecture and data choices. Manual inspection of 200 stratified errors reveals four dominant modes: (1) **Parse failures** (HARE: 36%, Vanilla: 59%), the largest source; (2) **Boundary confusion**: correct target, wrong severity ($\sim$18% vs. $\sim$12%); (3) **Multi-target omission**: HARE recovers 41% vs. 34% for Vanilla; (4) **Implicit hate via teencode/sarcasm**: harder for both models. Mode (4) includes cases where teencode substitution obscures offensive content (e.g., *"ngu vãi cả lòn"*, English: a vulgar expletive meaning "extremely stupid"), and sarcastic compliments that reverse literal meaning (e.g., *"gì hay thật đấy"*, English: "how wonderful," used sarcastically to demean). Both models assign *Normal* to such examples, suggesting that surface-token classifiers require explicit sarcasm-awareness mechanisms or augmented training data targeting these phenomena.

VII. ABLATION STUDY AND DISCUSSION

This section presents ablation experiments to isolate the contribution of rationale semantic structure from confounding factors, analyzes boundary redistribution mechanisms, and discusses threats to validity.

Multi-seed ablation controls. To disentangle rationale semantic content from extra gradient steps and training-distribution shifts, we ran two matched-budget controls on the 1,221-sample Stage 2 training subset across three seeds (42, 123, 456): (a) *Shuffled-Rationale*, where rationale steps are randomly permuted within each example, preserving text statistics but breaking semantic order; (b) *Plain-Continuation*, where rationale fields are removed, continuing only on comment text and labels, with Normal examples included proportional to training distribution to prevent class-distribution confounds. Table VI reports F1-Macro results.

Two findings emerge. First, **HARE marginally outperforms Shuffled-Rationale on average**: +1.15pp F1-Macro (seed 42), −0.07pp (seed 123), +2.25pp (seed 456), mean +1.11pp. The near-zero gap at seed 123 suggests the semantic-structure signal is modest, though the direction is positive on two of three seeds. Shuffled-Rationale preserves identical data selection, class distribution, and Hate/Offensive ratio (Δ=+26.8pp); only semantic coherence differs, providing a controlled contrast for rationale semantics. Second, **Plain-Continuation with Normal examples exceeds HARE** (mean 37.05% vs. 35.93% across all three seeds), indicating that the Stage 2 benefit conflates data exposure, class rebalancing, and rationale content. Additional training on the hateful subset, even without rationales, yields gains exceeding those of rationale-augmented training alone.

Why not adopt Plain-Continuation? (a) HARE marginally outperforms Shuffled-Rationale (+1.11pp mean F1-Macro), suggesting semantic structure contributes, though the effect is small and not consistent across all seeds; (b) Plain-Continuation includes Normal-class examples that the HARE CoT format structurally omits, introducing a class-distribution confound; (c) the HARE rationale structure enables future inference-time explanation. The HARE−Shuffled contrast remains the cleaner signal, even if the magnitude is modest.

A. Why Does Boundary Redistribution Occur?

The precision–recall pattern is most parsimoniously explained as prediction-threshold lowering for Offensive labels. Three mechanisms may underlie this shift: **(i)** the Hate-dominant rationale set (60.5%) may sharpen Hate prototypes, reclassifying marginal examples as Offensive; **(ii)** the fourth rationale step may transfer a finer boundary; **(iii)** 76.8% rejection may bias toward high-confidence patterns. The HARE $>$ Shuffled finding supports (ii) but cannot distinguish it from (i). This extends English-language findings on Offensive–Hate instability [19], [22] to Vietnamese.

B. Threats to Validity

The ablation cannot fully disentangle rationale semantics from hard-example mining (23% selection rate) and Hate-dominant class rebalancing (Δ=+26.8pp). The HARE–Shuffled gap is small (+1.11pp mean) with one seed reversal, limiting confidence in the semantic-structure claim. Stage 2 data overlap adds a memorization confound; Plain-Continuation outperforms HARE across all three seeds. Additional threats include same-family LLM verification, Religion-class scarcity ($<$30 test samples), and annotator homogeneity (Ho Chi Minh City university students, ages 20–22). At F1-Macro 35.93%, deployment without human-in-the-loop review is not recommended.

C. Constrained Decoding Experiment

To address parse failures without retraining, we tested constrained decoding via logit masking, restricting the model vocabulary to valid label tokens at each generation step. Table VII reports multi-seed results.

Constrained decoding collapses F1-Micro from 60%/57% to 21%/17% for Vanilla/HARE because naïve logit masking

Table VII
CONSTRAINED DECODING VS. FREE-TEXT GENERATION (%), MEAN OVER SEEDS {42,123,456}. CONSTRAINED DECODING CATASTROPHICALLY DEGRADES ALL METRICS.

	F1-Micro		F1-Macro	
Model	**Free**	**Constr.**	**Free**	**Constr.**
Vanilla	60.04	21.23	32.75	10.02
HARE	57.08	17.21	35.93	8.04

forces premature commitment in a variable label-order generation setting. Fixed-order retraining or grammar-constrained generation methods are needed but were beyond the scope of this work.

VIII. CONCLUSION

We presented a two-stage QLoRA pipeline adapting the HARE rationale-augmented training to Vietnamese targeted hate speech detection on ViTHSD. HARE achieves the best F1-Macro (35.93±0.63%, pooled bootstrap $p<0.001$) and the best parseable-only F1-Micro (46.71% vs. 44.58%, $p=0.004$); the aggregate Vanilla F1-Micro advantage is a parse-failure artifact. Per-label analysis reveals an asymmetric boundary shift (+6.84 avg F1 Offensive, +2.77 Hate), with gains overwhelmingly recall-driven. Multi-seed ablation shows a modest semantic-structure advantage over shuffled controls (+1.11pp mean), though Plain-Continuation achieves higher aggregate F1-Macro (37.05%), suggesting the Stage 2 benefit partly reflects data exposure alongside rationale content.

For Vietnamese content moderation, the recall advantage of HARE (+29pp Individuals#Offensive, +31pp Groups#Offensive) suits first-pass triage where missing implicit hate carries higher cost than false positives, conditional on reducing the 36% parse-failure rate. Differential parse-failure rates can reverse model rankings; we recommend routinely reporting parseable-only metrics.

Future work should prioritize: (1) constrained generation via retraining with a canonical label ordering to eliminate parse failures, (2) cross-family rationale verification using GPT-4o or Claude to address the same-family validation bias (Sec. IV: Gemini generates rationales and verifies label consistency), which may inflate the 23% selection rate, (3) extension to other Vietnamese hate speech datasets (ViHSD, ViHOS) to test generalizability, and (4) investigation of sarcasm- and teencode-aware augmentation strategies to address the implicit hate failures identified in Sec. VI-C. The boundary redistribution phenomenon may generalize to other low-resource tonal languages with severity-graded hate speech taxonomies, such as Thai or Cantonese, where cultural indirection similarly complicates automatic moderation.

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